

MANOUANE EST, QC, Canada

WMO: 715210

Lat: 50.656N

Lon: 70.532W

Elev: 1631

StdP: 13.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.6	-30.3	-42.6	0.5	-35.3	-37.1	0.7	-30.5	27.6	-3.1	25.3	-3.1	3.9	100	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	78.3	63.2	75.0	61.9	71.7	60.4	66.9	73.3	64.8	70.9	63.0	68.7	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	97.8	69.3	62.7	90.6	67.2	60.7	84.3	65.5	32.4	72.6	30.8	71.2	29.3	68.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	22.1	19.3	DB	-43.7	84.9	4.0	3.3	-46.6	87.3	-48.9	89.2	-51.2	91.0	-54.1	93.4	(4)
(5)			WB	-44.0	71.0	4.0	2.5	-46.9	72.8	-49.2	74.2	-51.5	75.7	-54.4	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.6	-3.6	-1.9	11.7	27.8	42.9	54.7	59.8	58.3	49.7	37.0	22.7	5.4		
	DBStd	24.68	14.87	14.38	14.22	10.01	8.14	7.26	5.01	5.29	7.14	7.19	9.94	13.85		
	HDD50	7948	1662	1453	1187	665	251	31	0	4	89	407	818	1381		
	HDD65	12609	2127	1873	1652	1115	687	320	174	219	461	867	1268	1846		
	CDD50	849	0	0	0	0	29	171	305	259	81	4	0	0		
	CDD65	37	0	0	0	0	1	9	14	10	3	0	0	0		
	CDH74	383	0	0	0	0	27	116	145	77	18	0	0	0		
	CDH80	47	0	0	0	0	4	17	17	7	2	0	0	0		
	WSAvg	7.6	7.6	7.6	8.0	8.1	7.0	7.2	7.0	7.0	7.7	8.6	8.5	7.2		
	PrecAvg	36.3	1.9	1.6	1.9	2.4	2.8	3.6	4.9	4.6	4.3	3.2	3.0	2.2		
(15) Precipitation	PrecMax	42.6	3.6	2.8	3.3	4.0	5.7	6.0	7.1	6.9	9.1	4.7	4.7	4.2		
	PrecMin	29.9	0.9	0.5	0.6	0.7	1.4	1.5	2.6	2.3	1.9	1.9	1.8	1.2		
	PrecStd	3.2	0.7	0.5	0.7	0.8	0.9	1.1	1.1	1.2	1.4	0.8	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	37.1	47.6	57.6	77.5	82.3	82.2	80.7	76.5	63.0	48.4	35.7		
		MCWB	36.1	33.8	38.3	43.2	58.1	64.8	66.2	67.3	65.8	55.0	46.9	35.2		
	2%	DB	31.8	32.3	40.5	51.3	70.5	77.3	78.1	75.9	70.2	56.1	42.0	32.8		
		MCWB	31.6	30.7	34.4	40.2	55.0	61.8	63.1	64.2	62.2	51.8	40.5	32.3		
	5%	DB	24.6	27.1	36.0	46.2	64.5	73.1	74.6	72.3	65.7	52.2	37.6	29.4		
		MCWB	23.8	25.4	32.3	38.0	51.3	59.1	62.1	62.5	59.5	48.8	36.2	29.0		
	10%	DB	19.1	21.6	32.6	42.0	59.1	68.9	71.1	68.7	61.8	48.1	34.9	25.5		
		MCWB	18.4	20.2	29.9	35.8	48.8	57.4	60.8	60.8	56.7	45.6	34.0	24.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	34.2	40.2	45.4	62.2	68.6	69.9	69.2	67.9	57.0	47.8	35.6		
		MCDB	35.3	35.2	45.1	53.7	70.6	77.0	76.8	75.8	72.5	60.8	48.9	35.8		
	2%	WB	30.1	30.3	35.8	41.5	56.7	65.0	66.4	66.6	64.1	53.2	41.0	32.3		
		MCDB	30.5	32.3	38.8	48.4	66.1	72.2	72.9	72.4	68.6	55.6	41.6	32.6		
	5%	WB	22.6	25.0	32.7	39.1	53.5	62.0	64.3	64.4	60.8	49.2	36.5	28.7		
		MCDB	23.1	26.9	35.5	45.1	62.7	68.9	71.0	69.4	64.9	51.2	37.0	29.1		
	10%	WB	17.5	20.1	29.7	36.7	50.1	59.4	62.6	62.3	57.6	45.5	33.9	24.8		
		MCDB	18.3	21.5	32.6	41.4	58.5	66.5	68.7	67.0	61.0	47.7	34.9	25.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	24.6	25.7	21.7	21.6	20.7	18.5	17.4	16.4	11.8	12.0	17.9		
		MCDBR	20.6	20.1	25.5	27.5	33.3	27.9	25.6	24.2	23.6	20.0	13.1	14.5		
		MCWBR	20.5	18.6	20.1	18.2	18.5	14.0	13.0	13.3	15.2	15.2	11.5	14.6		
	5% WB	MCDBR	21.0	19.0	22.2	22.7	28.5	22.6	20.3	19.2	20.4	17.3	11.7	14.8		
		MCWBR	20.9	17.5	18.3	15.7	17.6	13.2	11.9	12.0	14.9	14.4	11.3	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.252	0.281	0.322	0.340	0.368	0.380	0.359	0.342	0.310	0.278	0.240		
	taud		2.286	2.277	2.247	2.246	2.435	2.409	2.390	2.463	2.487	2.545	2.444	2.314		
	Ebn at Noon		257	284	294	292	291	282	276	278	271	262	241	239		
	Edh at Noon		21	27	35	39	34	36	36	31	28	22	18	17		
(44) All-Sky Solar Radiation	RadAvg		380	734	1164	1521	1708	1721	1559	1262	950	542	277	223		
	RadStd		45	64	103	115	125	139	137	77	72	54	30	32		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	+443	+472	N/A	N/A

Nomenclature: See separate page